Grade 11 Term 2 Home Practice – C3

Anatomy of a confidence interval

Global Economics – 2026–2027

Contents

Evaluated criteria 1

1 Introduction 1

2 Notation reference and practice procedure 3

3 General Worked Problems 4

4 Advanced Problems 6

Evaluated criteria

C3 Identifies the components of a confidence interval, including point estimate, critical value, standard error, and margin of error, explaining how each component affects precision and reliability.

How to practise. Try each task before opening its blue solution link. Use the arrows to return to the contents, the problem list or the current problem.

1 Introduction

Problems:

1.1. Key terms and highlight guide

1.2. From transport spending to an interval

1.3. Quick check: centre and width

1.1 Key terms and highlight guide

A confidence interval combines a sample-based point estimate with a margin of error to estimate a population mean. Use the following table as a guide to the components, symbols, and highlights used throughout this document.
Greens connect sample quantities, and reds connect population quantities. Two blue shades distinguish the confidence level (central area) from its critical value. Separate warm shades distinguish standard error, margin of error, endpoints, and full width; names and symbols identify each quantity even without color.

Terms and symbols	Meaning in context
Population mean, μ	The fixed population parameter being estimated.
Point estimate, $\bar{x}$	Sample mean and interval centre; for example, 60 units.
Standard error, SE	Spread of sample means across repeated samples, or its estimate; for example, 2 units. Distinct from individual-observation spread.
Margin of error, MoE	Distance from centre to an endpoint: MoE $= z^*$ SE. For example, 1.96 (2) = 3.92 units.
Endpoints, $\bar{x} \pm$ MoE	Lower and upper bounds: 60 $\pm$ 3.92 gives 56.08 and 63.92 units.
Full width, 2 MoE	Distance between endpoints, twice the margin of error; here 2(3.92) = 7.84 units.
Confidence level, C	Nominal long-run coverage under the model; 95% = 0.95 is the central normal area.
Standard normal critical value, z^*	Unitless multiplier selected by C; 1.96 bounds the central area 0.95 between $-z^*$ and z^*.



1.2 From transport spending to an interval

A survey estimates mean weekly household transport spending at 60 thousand COP. A margin of error of 5 thousand COP gives an interval from 60 − 5 = 55 to 60 + 5 = 65 thousand COP, centred at 60 thousand COP.



1.3 Quick check: centre and width

If the standard error decreases while the point estimate and confidence level are held fixed, what happens to the centre and the full width? Explain your prediction using the formulas below.



2 Notation reference and practice procedure

This section develops the components introduced in the opening guide through detailed notation and practice steps. Here MoE denotes the margin of error; C denotes the confidence level as a decimal.

$$\text{MoE} = z^* \times \text{SE}, \qquad \text{full width} = 2\,\text{MoE},$$
$$\text{interval} = \bar{x} \pm \text{MoE} = [\,\bar{x} - \text{MoE},\ \bar{x} + \text{MoE}\,].$$

2.1 Notation reference

Point estimate $\bar{x}$**.** The sample mean. It estimates the population mean and sets the centre.

Critical value z^***.** The unitless multiplier for the chosen confidence level. That level is the area under the standard normal curve from $-z^*$ to z^*.

Standard error SE**.** The standard deviation of the sample mean across repeated samples, or an estimate of that deviation.

Margin of error MoE**.** The distance from the estimate to either endpoint. Smaller MoE gives a narrower, more precise interval.

2.2 Practice procedure

1. Identify the target mean, sample mean, confidence level, matching z^* and supplied SE.
2. Find MoE. Subtract and add MoE to find the endpoints. Check that the estimate, SE, MoE, endpoints and width share the same units.
3. Compare estimates for the centres, products z^* SE for the widths, and confidence levels for nominal long-run coverage. Under the sampling and normal-model assumptions, a 95% procedure captures the fixed population mean in about 95% of repeated samples.

3 General Worked Problems

Problems:

3.1. Household grocery spending (C3)
3.2. Two estimates of average freight cost (C3)

3.1 Household grocery spending (C3)

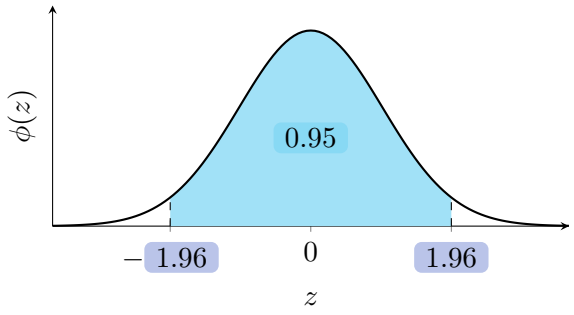
An analyst estimates the mean weekly grocery bill for households in a district. The random sample mean is 84 thousand COP and the standard error of the sample mean is 3 thousand COP. A two-sided 95% normal interval uses $z^* = 1.96$. A two-sided 90% interval uses $z^* = 1.645$.

Questions/Tasks

C3 Write the interval in the form estimate ± critical value × standard error. Identify all four components, including the margin of error, and explain what each contributes. Sketch the standard normal curve and mark the central area associated with the critical value.

C3 Recalculate the margin of error for 90% confidence using the same data. Explain what happens to the centre, width and long-run coverage.

C3 – Identify the components and their roles



The **point estimate** of 84 thousand COP sets the centre. The **standard error** of 3 thousand COP measures how much the sample mean typically varies across samples. The unitless **critical value** 1.96 gives a central area of about 0.95, with about 0.025 in each tail.

The **margin of error** is MoE = 1.96(3) = 5.88 thousand COP, the distance from the centre to each endpoint.

The 95% interval for the district's mean weekly grocery bill is

$$84 \pm 1.96(3) = [84 - 5.88,\ 84 + 5.88] = [78.12,\ 89.88] \quad \text{thousand COP.}$$

C3 – Separate confidence from precision

With the same sample, the centre stays at 84 thousand COP and SE stays at 3 thousand COP. At 90% confidence, the smaller critical value gives MoE = 1.645(3) = 4.935 thousand COP and the interval $84 \pm 4.935 = [79.065, 88.935]$ thousand COP. The width drops from $2(5.88) = 11.76$ to $2(4.935) = 9.87$ thousand COP. The narrower interval gives a more precise estimate of the mean grocery bill, while the procedure's nominal long-run coverage drops from 95% to 90%.

3.2 Two estimates of average freight cost (C3)

Two simple random samples estimate the mean freight cost per container on a route. Both sample means are 250 USD. Sample A gives SE = 8 USD and sample B gives SE = 4 USD. Both use the two-sided critical value $z^* = 1.96$ for 95% confidence.

Questions/Tasks



C3 Find both margins and widths. Identify which components stay fixed, and explain whether the smaller standard error changes nominal coverage.

C3 Suppose sample B's point estimate had been 254 USD and its SE still 4 USD. Give the new interval and describe how it changes.

C3 – Explain the effect of a smaller standard error

Both samples have a point estimate of 250 USD and a critical value of 1.96. All amounts below are in USD per container.

A: $\text{MoE}_A = 1.96\,(\,8\,) = 15.68$, $250 \pm 15.68 = [\,234.32,\ 265.68\,]$, $2\,\text{MoE}_A = 31.36$.

B: $\text{MoE}_B = 1.96\,(\,4\,) = 7.84$, $250 \pm 7.84 = [\,242.16,\ 257.84\,]$, $2\,\text{MoE}_B = 15.68$.

Halving the standard error halves the margin and width. Sample B gives a more precise interval for the mean freight cost, with the same centre. Both procedures target 95% long-run coverage under their sampling and model assumptions.



C3 – Explain the effect of moving the point estimate

The same z^* and SE give MoE $= 1.96\,(\,4\,) = 7.84$ USD. With the higher estimated mean freight cost, the new interval is $254 \pm 7.84 = [\,246.16,\ 261.84\,]$ USD per container. Both endpoints move up by 4 USD with the centre. The width stays at 15.68 USD, so precision is unchanged, and the procedure still targets 95% coverage.



4 Advanced Problems

Problems:

4.1. Monthly electricity costs for small shops (C3)

4.1 Monthly electricity costs for small shops (C3)

A business association uses two random samples to estimate the same population mean, the monthly electricity cost per small shop in a town. Both samples give a mean of 120 thousand COP. For their two-sided normal intervals, the analysts report the following inputs.

Sample	Confidence level	Critical value z^*	SE (thousand COP)
A	95%	1.96	10
B	99%	2.576	5

Questions/Tasks

C3 Predict which interval will be narrower. Calculate both margins of error, intervals and full widths. Use z^*SE to explain how B's smaller standard error and larger critical value work against each other. Compare the centres, precision and nominal long-run coverage.

C3 – Compare intervals when two inputs change

Predict that B will be narrower. Its standard error is half as large, while its critical value is less than twice as large. Since MoE $= z^* \times$ SE, the product for B is smaller. All amounts below are in thousand COP per shop per month.

A: $\text{MoE}_A = 1.96(10) = 19.60$, $120 \pm 19.60 = [100.40, 139.60]$, $2\,\text{MoE}_A = 39.20$.

B: $\text{MoE}_B = 2.576(5) = 12.88$, $120 \pm 12.88 = [107.12, 132.88]$, $2\,\text{MoE}_B = 25.76$.

Both intervals are centred at 120 thousand COP. B gives a more precise estimate of the mean monthly electricity cost because its interval is narrower. Coverage describes how often the procedure captures the population mean across repeated samples. Under the sampling and normal-model assumptions, B targets 99% and A targets 95%.